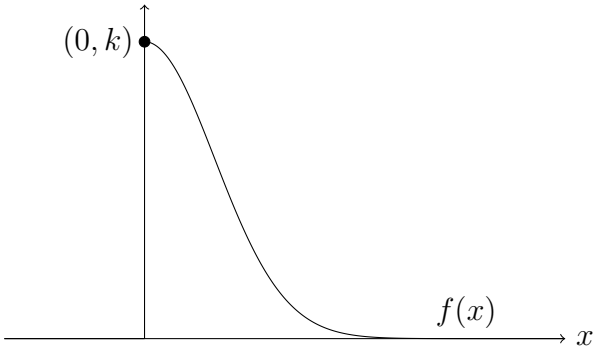
(i)



(ii) By the normal distribution formula, $\sqrt{\frac{2}{\pi}}e^{-2x^2}$ is the density of $N\left(0, \frac{1}{4}\right)$

$$\therefore \int_{-\infty}^{\infty} \sqrt{\frac{2}{\pi}}e^{-2x^2}\,dx = 1$$

$$\implies \int_{0}^{\infty} e^{-2x^2}\,dx = \frac{1}{2}\sqrt{\frac{\pi}{2}}$$

$$\because \int_{0}^{\infty} ke^{-2x^2}\,dx = 1$$

$$\therefore \frac{k}{2}\sqrt{\frac{\pi}{2}} = 1 \implies k = 2\sqrt{\frac{2}{\pi}}$$

(iii)

$$\begin{aligned} E(X) &= \int_{0}^{\infty} xke^{-2x^2}\,dx \\ &= 2\sqrt{\frac{2}{\pi}}\left[-\frac{1}{4}e^{-2x^2}\right]_{0}^{\infty} \\ &= \frac{1}{\sqrt{2\pi}} \end{aligned}$$

$$\begin{aligned} \int_{0}^{\infty} x^2e^{-2x^2}\,dx &= \int_{0}^{\infty} x \cdot xe^{-2x^2}\,dx \\ &= \left[-\frac{1}{4}xe^{-2x^2}\right]_{0}^{\infty} + \frac{1}{4}\int_{0}^{\infty} e^{-2x^2}\,dx \\ &= \frac{1}{4} \cdot \frac{1}{2}\sqrt{\frac{\pi}{2}} \\ &= \frac{\sqrt{\pi}}{8\sqrt{2}} \end{aligned}$$

$$\begin{aligned} E(X^2) &= k\int_0^{\infty} x^2 e^{-2x^2}\,dx \\ &= 2\sqrt{\frac{2}{\pi}}\cdot\frac{\sqrt{\pi}}{8\sqrt{2}} \\ &= \frac{1}{4} \end{aligned}$$

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2$$

$$\mathrm{Var}(X) = \frac{1}{4} - \frac{1}{2\pi}$$

(iv)
Let the median be m.

$$\frac{1}{2} = \int_0^m \frac{4}{\sqrt{2\pi}} e^{-2x^2}\,dx$$

Put $t = 2x$,

$$dx = \frac{1}{2}\,dt$$

$$\begin{aligned} \frac{1}{2} &= \frac{4}{\sqrt{2\pi}}\int_0^m e^{-2x^2}\,dx \\ &= \frac{4}{\sqrt{2\pi}}\int_0^{2m} e^{-\frac{1}{2}t^2}\cdot\frac{1}{2}\,dt \\ &= 2\left(\frac{1}{\sqrt{2\pi}}\int_0^{2m} e^{-\frac{1}{2}t^2}\,dt\right) \\ &= 2\left(\Phi(2m) - \frac{1}{2}\right) \end{aligned}$$

$$\Phi(2m) = \frac{3}{4}$$

$$2m = 0.6745 \implies m = 0.337 \quad (3\ s.f.)$$

This solution was submitted by **Jon**
Published and refined by **Jon**

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